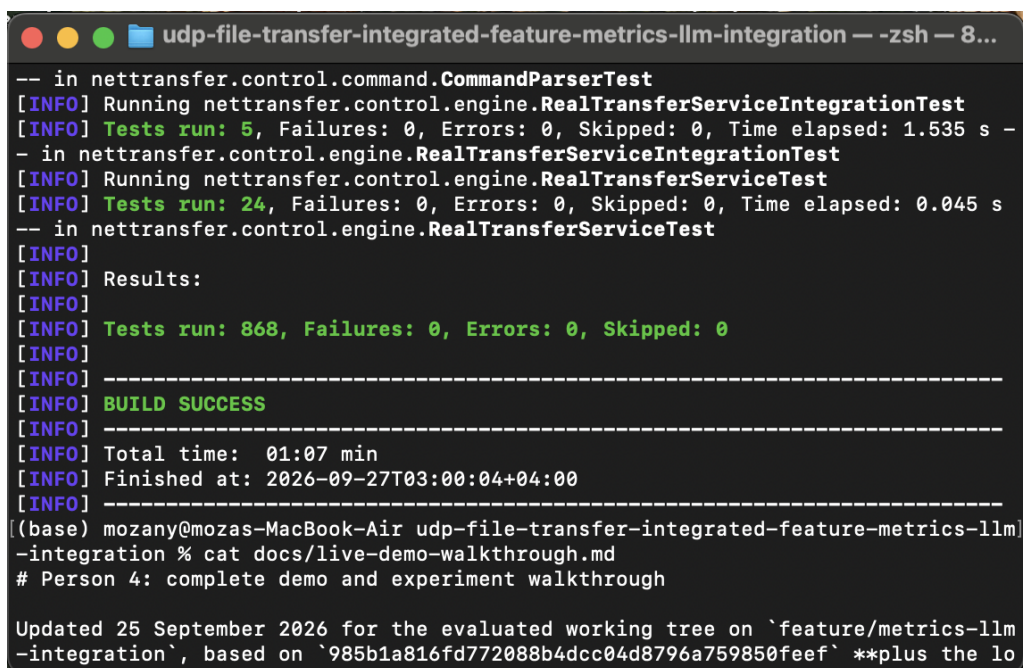


Experimental Evaluation Evidence

This document provides supporting evidence for the experimental evaluation of the reliable UDP file-transfer system. It documents the environment validation, baseline, 2% packet-loss and 200 ms delay experiments for both file sizes, together with the additional window-size comparison used to evaluate performance under increased network latency.

A terminal window titled 'udp-file-transfer-integrated-feature-metrics-llm-integration --zsh -- 8...' displays the output of a Maven test execution. The output shows three test phases: 'nettransfer.control.command.CommandParserTest' (5 tests, 1.535s), 'nettransfer.control.engine.RealTransferServiceIntegrationTest' (24 tests, 0.045s), and 'nettransfer.control.engine.RealTransferServiceTest' (868 tests, 0.045s). All tests passed with zero failures or errors. The summary indicates 'BUILD SUCCESS' with a total time of 01:07 min, finished at 2026-09-27T03:00:04+04:00. The user then runs 'cat docs/live-demo-walkthrough.md' and reads the first line: '# Person 4: complete demo and experiment walkthrough'. The terminal also shows a commit hash and a timestamp: 'Updated 25 September 2026 for the evaluated working tree on `feature/metrics-llm-integration`, based on `985b1a816fd772088b4dcc04d8796a759850feef` **plus the lo'.

```
-- in nettransfer.control.command.CommandParserTest
[INFO] Running nettransfer.control.engine.RealTransferServiceIntegrationTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.535 s -
- in nettransfer.control.engine.RealTransferServiceIntegrationTest
[INFO] Running nettransfer.control.engine.RealTransferServiceTest
[INFO] Tests run: 24, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.045 s
-- in nettransfer.control.engine.RealTransferServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 868, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:07 min
[INFO] Finished at: 2026-09-27T03:00:04+04:00
[INFO] -----
((base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm
-integration % cat docs/live-demo-walkthrough.md
# Person 4: complete demo and experiment walkthrough

Updated 25 September 2026 for the evaluated working tree on `feature/metrics-llm
-integration`, based on `985b1a816fd772088b4dcc04d8796a759850feef` **plus the lo
```

Figure 1: Maven Test Execution

Before conducting the experiments, the integrated implementation was validated using the automated test suite. All 868 tests completed successfully with no failures or errors, confirming that the implementation was stable before evaluation.

```
udp-file-transfer-integrated-feature-metrics-llm-integration — zsh — 8...
[WARNING] - META-INF/versions/9/module-info
[WARNING] error_prone_annotations-2.27.0.jar, gson-2.11.0.jar, udp-file-transfer
.jar define 1 overlapping resource:
[WARNING] - META-INF/MANIFEST.MF
[WARNING] maven-shade-plugin has detected that some files are
[WARNING] present in two or more JARs. When this happens, only one
[WARNING] single version of the file is copied to the uber jar.
[WARNING] Usually this is not harmful and you can skip these warnings,
[WARNING] otherwise try to manually exclude artifacts based on
[WARNING] mvn dependency:tree -Ddetail=true and the above output.
[WARNING] See https://maven.apache.org/plugins/maven-shade-plugin/
[INFO] Replacing original artifact with shaded artifact.
[INFO] Replacing /Users/mozany/Downloads/udp-file-transfer-integrated-feature-me
trics-llm-integration/target/udp-file-transfer.jar with /Users/mozany/Downloads/
udp-file-transfer-integrated-feature-metrics-llm-integration/target/udp-file-tra
nsfer-1.0.0-shaded.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 16.959 s
[INFO] Finished at: 2026-09-27T03:05:13+04:00
[INFO] -----
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm
-integration %
```

Figure 2: Maven Build Success

Following testing, the project was packaged successfully into the executable JAR used for the experimental runs.

```
udp-file-transfer-integrated-feature-metrics-llm-integration — zsh — 8...

    with open(path, "wb") as f:
        f.write(data)

    print(f"CREATED: {path}")
    print(f"SIZE: {len(data)} bytes")
    print(f"SHA256: {hashlib.sha256(data).hexdigest()}")
    print()
PY
CREATED: storage/outgoing/person4-small.bin
SIZE: 262267 bytes
SHA256: 873b22f9250a56ebc54af3b98f63d688608d1207d3262a89f08ce52ce13245f4

CREATED: storage/outgoing/person4-large.bin
SIZE: 2097275 bytes
SHA256: f098a5e9cddd0d6836e447018f237d765d65d8c524377fd943d49063b26304c1

(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm
-integration % ls -lh storage/outgoing/person4-*.bin
-rw-r--r--  1 mozany  staff   2.0M Sep 27 03:03 storage/outgoing/person4-large.b
in
-rw-r--r--  1 mozany  staff  256K Sep 27 03:03 storage/outgoing/person4-small.b
in
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm
-integration %
```

Figure 3: Test File Generation

Two test files were generated for evaluation: a 262,267-byte small file and a 2,097,275-byte large file. The same files were retained across the different network conditions to maintain consistency between experiments.

Baseline Experiments

```
person4-small-baseline-20260927-030640 --zsh -- 80x24
javac 27
(base) mozany@mozas-MacBook-Air person4-small-baseline-20260927-030640 % java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=baseline \
-Dnettransfer.logsRoot="$RUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] baseline: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per
direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-featu
re-metrics-llm-integration/target/evaluation/person4-small-baseline-20260927-030
640/logs/standalone/run-41eb09cc-0491-4ba5-a5a5-15d91579ae9a/events-receiver.js
on
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-me
trics-llm-integration/target/evaluation/person4-small-baseline-20260927-030640/s
torage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-small-baseline-20260927-030640 %

mozany — java -Dnettransfer.impairment.enabled=true -Dnettransfer.im...
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout:
500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:09:12.584708Z
transfer_id=09e1504a-ae43-48d8-aa9c-1aed2fd8a7d1; run_id=09e1504a-ae43-48d8-aa9c
-1aed2fd8a7d1; protocol_transfer_id=aed64735-ced3-45c0-94fb-0d892fc09621
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_final
ity=FINAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=0.138240792
sender_ack_based_rate_mbps=15.177401472063325
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=2.934768155642023
rtt_p95_ms=7.1195
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>
```

Figure 4: Small-File Baseline Configuration

The evaluation began with the small file under baseline conditions, with no intentional packet loss or delay. An 8-packet (8192-byte) window and a 500 ms timeout were used.

```
person4-large-baseline-20260927-031446 --zsh -- 80x24
java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=baseline \
-Dnettransfer.logsRoot="$RUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] baseline: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per
direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-featu
re-metrics-llm-integration/target/evaluation/person4-large-baseline-20260927-031
446/logs/standalone/run-a3fc2028-8e51-48eb-9867-e767e2130a16/events-receiver.js
on
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-me
trics-llm-integration/target/evaluation/person4-large-baseline-20260927-031446/s
torage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-large-baseline-20260927-031446 %

person4-small-baseline-20260927-030640 — java -Dnettransfer.impair...
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout:
500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:18:20.732399Z
transfer_id=1809ef42-efec-4a34-81f4-173dd178e9aa; run_id=1809ef42-efec-4a34-81f4
-173dd178e9aa; protocol_transfer_id=4b70c281-a575-41a5-9aee-bac0835aa7c5
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_final
ity=FINAL; provenance=REAL
file_size_bytes=2097275
sender_confirmed_acked_payload_bytes=2097275
sender_ack_progress_percent=100
live_elapsed_time_sec=0.444889125
sender_ack_based_rate_mbps=37.713216748105495
packets_sent=2049
retransmissions=0
acks_received=2049
distinct_packets_acked=2049
packets_timed_out=0
rtt_sample_count=2049
rtt_mean_ms=1.1954251464128853
rtt_p95_ms=3.517125
endpoint_local_udp_payload_bytes_emitted=2152935
endpoint_emission_accounting_complete=true
transfer>
```

Figure 5: Small-File Baseline Transfer

```
person4-large-baseline-20260927-031446 --zsh -- 82x27
=== FINAL RECONCILED METRICS ===
file_size_bytes: 2097275
payload_bytes_delivered: 2097275
transfer_time_sec: 0.444889125
throughput_mbps: 37.713216748105495
transfer_success: True
integrity_verified: True
packets_sent: 2049
packets_received: 2049
acks_received: 2049
packets_timed_out: 0
packets_duplicated: 0
packets_dropped: 0
retransmissions: 0
retransmission_ratio: 0.0
protocol_overhead_bytes: 111231
protocol_overhead_ratio: 0.050364816758478356
rtt_mean_ms: 1.1954251464128853
rtt_p95_ms: 3.517125
window_bytes_requested: 8192
window_packets: 8
timeout_ms: 500
packet_loss_rate: 0.0
delay_ms: 0.0
scenario: baseline
(base) mozany@mozas-MacBook-Air person4-large-baseline-20260927-031446 %

person4-small-baseline-20260927-030640 — java -Dnettransfer.impairment.e...
Start accepted [REAL]
transfer_id=1809ef42-efec-4a34-81f4-173dd178e9aa; run_id=1809ef42-efec-4a34-81f4-173dd
178e9aa; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 m
s; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:18:20.732399Z
transfer_id=1809ef42-efec-4a34-81f4-173dd178e9aa; run_id=1809ef42-efec-4a34-81f4-173dd
178e9aa; protocol_transfer_id=4b70c281-a575-41a5-9aee-bac0835aa7c5
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FI
NAL; provenance=REAL
file_size_bytes=2097275
sender_confirmed_acked_payload_bytes=2097275
sender_ack_progress_percent=100
live_elapsed_time_sec=0.444889125
sender_ack_based_rate_mbps=37.713216748105495
packets_sent=2049
retransmissions=0
acks_received=2049
distinct_packets_acked=2049
packets_timed_out=0
rtt_sample_count=2049
rtt_mean_ms=1.1954251464128853
rtt_p95_ms=3.517125
endpoint_local_udp_payload_bytes_emitted=2152935
endpoint_emission_accounting_complete=true
transfer>
```

Figure 6: Small-File Baseline Results


```
person4-small-baseline-20260927-030640 — zsh — 80x24
javac 27
(base) mozany@mozas-MacBook-Air person4-small-baseline-20260927-030640 % java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=baseline \
-Dnettransfer.logsRoot="$SRUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] baseline: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per di
rection, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-featu
re-metrics-llm-integration/target/evaluation/person4-small-baseline-20260927-030
640/logs/standalone/run-41eb09cc-0491-4ba5-a5a5-15d91679ae9a/events-receiver.js
on
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-me
trics-llm-integration/target/evaluation/person4-small-baseline-20260927-030640/s
torage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-small-baseline-20260927-030640 %

mozany — java -Dnettransfer.impairment.enabled=true -Dnettransfer.im...
Simulator baseline: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per di
rection, seed=42; START/FINISH unaffected.
Evaluation records: /Users/mozany/Downloads/udp-file-transfer-integrated-feature
-metrics-llm-integration/target/evaluation/person4-small-baseline-20260927-03064
0/target/evaluation/llm-2026-09-26T23-08-43.801821Z-3403230809502825959
Recorded model responses are untrusted evidence and may include answers rejected
by Java.
UDP transfer console. Type help for commands; natural-language requests use the
configured interpreter.
transfer> catalog
Approved files (paths relative to /Users/mozany/Downloads/udp-file-transfer-inte
grated-feature-metrics-llm-integration/target/evaluation/person4-small-baseline-
20260927-030640):
  sample = storage/outgoing/sample.bin
Approved receivers:
  receiver-a = 127.0.0.1:9000
transfer> start_transfer {"file_id":"sample","receiver_id":"receiver-a","window_
bytes":8192,"timeout_ms":500}
Start accepted [REAL]
transfer_id=09e1504a-ae43-48d8-aa9c-1aed2fd8a7d1; run_id=09e1504a-ae43-48d8-aa9c
-1aed2fd8a7d1; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout:
500 ms; retry limit: 5 consecutive rounds without progress
transfer>
```

Figure 7: Small-File Baseline Reconciliation

The small-file baseline transfer completed successfully with integrity verified and no packet drops, retransmissions, or timeouts. The reconciled throughput was 15.18 Mbps.

The same baseline configuration was then applied to the large file to evaluate performance as the transfer size increased.

```
person4-large-baseline-20260927-031446 — zsh — 80x24
java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=baseline \
-Dnettransfer.logsRoot="$SRUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] baseline: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per
direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-featu
re-metrics-llm-integration/target/evaluation/person4-large-baseline-20260927-031
446/logs/standalone/run-a3fc2028-8e51-48eb-9867-e767e2130a16/events-receiver.js
on
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-me
trics-llm-integration/target/evaluation/person4-large-baseline-20260927-031446/s
torage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-large-baseline-20260927-031446 %

person4-small-baseline-20260927-030640 — java -Dnettransfer.impair...
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=baseline \
-Dnettransfer.evaluation.record=true \
-cp "$JAR" nettransfer.cli.TransferCliMain \
"$SRUNROOT" \
['sample=storage/outgoing/sample.bin']
Simulator baseline: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per di
rection, seed=42; START/FINISH unaffected.
Evaluation records: /Users/mozany/Downloads/udp-file-transfer-integrated-feature
-metrics-llm-integration/target/evaluation/person4-large-baseline-20260927-03144
6/target/evaluation/llm-2026-09-26T23-18-11.145547Z-15792819285285681368
Recorded model responses are untrusted evidence and may include answers rejected
by Java.
UDP transfer console. Type help for commands; natural-language requests use the
configured interpreter.
transfer> start_transfer {"file_id":"sample","receiver_id":"receiver-a","window_
bytes":8192,"timeout_ms":500}
Start accepted [REAL]
transfer_id=1809ef42-efec-4a34-81f4-173dd178e9aa; run_id=1809ef42-efec-4a34-81f4
-173dd178e9aa; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout:
500 ms; retry limit: 5 consecutive rounds without progress
transfer>
```

Figure 8: Large-File Baseline Configuration

```
udp-file-transfer-integrated-feature-metrics-llm-integration — java -Dne...
-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-l
arge-baseline-20260927-031446/storage/outgoing/sample.bin
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm
-integration % cd "$SRUNROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=baseline \
-Dnettransfer.logsRoot="$SRUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] baseline: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per
direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-featu
re-metrics-llm-integration/target/evaluation/person4-large-baseline-20260927-031
446/logs/standalone/run-a3fc2028-8e51-48eb-9867-e767e2130a16/events-receiver.js
on
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-me
trics-llm-integration/target/evaluation/person4-large-baseline-20260927-031446/s
torage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-large-baseline-20260927-031446 %

person4-small-baseline-20260927-030640 — java -Dnettransfer.impair...
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=baseline \
-Dnettransfer.evaluation.record=true \
-cp "$JAR" nettransfer.cli.TransferCliMain \
"$SRUNROOT" \
['sample=storage/outgoing/sample.bin']
Simulator baseline: receiver DATA loss=0.0%, DATA/ACK delivery delay=0 ms per di
rection, seed=42; START/FINISH unaffected.
Evaluation records: /Users/mozany/Downloads/udp-file-transfer-integrated-feature
-metrics-llm-integration/target/evaluation/person4-large-baseline-20260927-03144
6/target/evaluation/llm-2026-09-26T23-18-11.145547Z-15792819285285681368
Recorded model responses are untrusted evidence and may include answers rejected
by Java.
UDP transfer console. Type help for commands; natural-language requests use the
configured interpreter.
transfer> start_transfer {"file_id":"sample","receiver_id":"receiver-a","window_
bytes":8192,"timeout_ms":500}
Start accepted [REAL]
transfer_id=1809ef42-efec-4a34-81f4-173dd178e9aa; run_id=1809ef42-efec-4a34-81f4
-173dd178e9aa; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout:
500 ms; retry limit: 5 consecutive rounds without progress
transfer>
```

Figure 9: Large-File Baseline Transfer

```
person4-small-baseline-20260927-030640 — -zsh — 80x24
throughput_mbps: 15.177401472863325
transfer_success: True
integrity_verified: True
packets_sent: 257
packets_received: 257
acks_received: 257
packets_acked: 257
packets_timed_out: 0
packets_duplicated: 0
packets_dropped: 0
retransmissions: 0
retransmission_ratio: 0.0
protocol_overhead_bytes: 14462
protocol_overhead_ratio: 0.052260514799677665
rtt_sample_count: 257
rtt_mean_ms: 2.934768155642023
rtt_p95_ms: 7.1195
window_bytes_requested: 8192
window_packets: 8
timeout_ms: 500
packet_loss_rate: 0.0
delay_ms: 0.0
scenario: baseline
(base) mozany@mozas-MacBook-Air person4-small-baseline-20260927-030640 %

mozany — java -Dnettransfer.impairment.enabled=true -Dnettransfer.im...
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout:
500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:09:12.584708Z
transfer_id=09e1504a-ae43-48d8-aa9c-1aed2fd8a7d1; run_id=09e1504a-ae43-48d8-aa9c
-1aed2fd8a7d1; protocol_transfer_id=aed64735-ced3-45c8-94fb-0d892fc09621
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_final
ity=FINAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=0.138240792
sender_ack_based_rate_mbps=15.177401472863325
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=2.934768155642023
rtt_p95_ms=7.1195
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>
```

Figure 10: Large-File Baseline Results

```
person4-small-baseline-20260927-030640 — -zsh — 80x24
echo "RECEIVER: $RECEIVER_DIR"

java -jar "$JAR" reconcile "$SENDER_DIR" "$RECEIVER_DIR"
SENDER: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm
-integration/target/evaluation/person4-small-baseline-20260927-030640/logs/stand
alone/09e1504a-ae43-48d8-aa9c-1aed2fd8a7d1
RECEIVER: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-ll
m-integration/target/evaluation/person4-small-baseline-20260927-030640/logs/stand
alone/run-41eb09cc-0491-4ba5-a5a5-15d91579ae9a
WARNING: Final field chunkSizeBytes in class nettransfer.metrics.TransferConfigu
ration has been mutated reflectively by class com.google.gson.internal.bind.Refle
ctiveTypeAdapterFactory$2 in unnamed module @7bfc12c (file:/Users/mozany/Downl
oads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/udp-fil
e-transfer.jar)
WARNING: Use --enable-final-field-mutation=ALL-UNNAMED to avoid a warning
WARNING: Mutating final fields will be blocked in a future release unless final
field mutation is enabled
[metrics] Summary: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-
metrics-llm-integration/target/evaluation/person4-small-baseline-20260927-030640
/logs/standalone/reconciled/aed64735-ced3-45c8-94fb-0d892fc09621/summary.json1
[metrics] Manifest: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-
metrics-llm-integration/target/evaluation/person4-small-baseline-20260927-03064
0/logs/standalone/reconciled/aed64735-ced3-45c8-94fb-0d892fc09621/manifest.json
(base) mozany@mozas-MacBook-Air person4-small-baseline-20260927-030640 %

mozany — java -Dnettransfer.impairment.enabled=true -Dnettransfer.im...
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout:
500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:09:12.584708Z
transfer_id=09e1504a-ae43-48d8-aa9c-1aed2fd8a7d1; run_id=09e1504a-ae43-48d8-aa9c
-1aed2fd8a7d1; protocol_transfer_id=aed64735-ced3-45c8-94fb-0d892fc09621
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_final
ity=FINAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=0.138240792
sender_ack_based_rate_mbps=15.177401472863325
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=2.934768155642023
rtt_p95_ms=7.1195
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>
```

Figure 11: Large-File Baseline Reconciliation

The transfer completed without packet loss, retransmissions, or timeouts, achieving a reconciled throughput of 37.71 Mbps.

Packet-Loss Experiments

```
person4-small-loss-2-20260927-032128 — -zsh — 82x27
NEW RUN: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm
integration/target/evaluation/person4-small-loss-2-20260927-032128
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm
integration % cd "$RUNROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=2 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=loss-2 \
-Dnettransfer.logsRoot="$RUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] loss-2: receiver DATA loss=2.0%, DATA/ACK delivery delay=0 ms per di
ection, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-featur
-metrics-llm-integration/target/evaluation/person4-small-loss-2-20260927-032128/l
gs/standalone/run-2e2a6ada-72c7-471d-8d19-da7ce5f57c42/events-receiver.json1
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-met
ics-llm-integration/target/evaluation/person4-small-loss-2-20260927-032128/stora
/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-small-loss-2-20260927-032128 %

person4-large-baseline-20260927-031446 — java -Dnettransfer.impairment.e...
Start accepted [REAL]
transfer_id=505a6f08-9d56-4d26-8282-855a530ac537; run_id=505a6f08-9d56-4d26-8282-855a5
30ac537; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 m
s; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:22:27.063852Z
transfer_id=505a6f08-9d56-4d26-8282-855a530ac537; run_id=505a6f08-9d56-4d26-8282-855a5
30ac537; protocol_transfer_id=c6c725ed-cd53-4f9f-9cb0-72331321434f
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FI
NAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=1.148090958
sender_ack_based_rate_mbps=1.827499803373593
packets_sent=273
retransmissions=16
acks_received=257
distinct_packets_acked=257
packets_timed_out=2
rtt_sample_count=241
rtt_mean_ms=2.756125850622406
rtt_p95_ms=5.905042
endpoint_local_udp_payload_bytes_emitted=286358
endpoint_emission_accounting_complete=true
transfer>
```

Figure 12: Small-File Loss Configuration


```
person4-small-loss-2-20260927-032128 -- -zsh -- 82x27
etrics-llm-integration/target/evaluation/person4-small-loss-2-20260927-032128/logs
/standalone/reconciled/c6c725ed-cd53-4f9f-9cb0-72331321434f/manifest.json
file_size_bytes: 262267
payload_bytes_delivered: 262267
transfer_time_sec: 1.148090958
throughput_mbps: 1.827499803373593
transfer_success: True
integrity_verified: True
packets_sent: 273
packets_received: 271
acks_received: 267
packets_timed_out: 2
packets_duplicated: 0
packets_dropped: 2
retransmissions: 16
retransmission_ratio: 0.05860805860805861
protocol_overhead_bytes: 31278
protocol_overhead_ratio: 0.10655265802517501
rtt_mean_ms: 2.756125850622406
rtt_p95_ms: 5.905042
window_bytes_requested: 8192
window_packets: 8
timeout_ms: 500
packet_loss_rate: 2.0
delay_ms: 0.0
scenario: loss-2
(base) mozany@mozas-MacBook-Air person4-small-loss-2-20260927-032128 %

person4-large-baseline-20260927-031446 -- java -Dnettransfer.impairment.e...
start accepted [REAL]
ransfer_id=505a6f08-9d56-4d26-8282-855a530ac537; run_id=505a6f08-9d56-4d26-8282-855a5
0ac537; protocol_transfer_id=unavailable
equested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 m
; retry limit: 5 consecutive rounds without progress
ransfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:22:27.063852Z
ransfer_id=505a6f08-9d56-4d26-8282-855a530ac537; run_id=505a6f08-9d56-4d26-8282-855a5
0ac537; protocol_transfer_id=c6c725ed-cd53-4f9f-9cb0-72331321434f
ngine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FI
AL; provenance=REAL
ile_size_bytes=262267
ender_confirmed_ackd_payload_bytes=262267
ender_ack_progress_percent=100
ive_elapsed_time_sec=1.148090958
ender_ack_based_rate_mbps=1.827499803373593
ackets_sent=273
etransmissions=16
cks_received=267
istinct_packets_ackd=267
ackets_timed_out=2
tt_sample_count=241
tt_mean_ms=2.756125850622406
tt_p95_ms=5.905042
ndpoint_local_udp_payload_bytes_emitted=286358
ndpoint_emission_accounting_complete=true
ransfer>
```

Figure 13: Small-File Loss Results

A 2% DATA packet-loss impairment was then introduced while retaining the 8192-byte window and 500 ms timeout. The small-file transfer remained successful, but required 16 retransmissions and experienced two recorded packet drops.

The 2% loss condition was then repeated using the large file to observe the effect of packet loss over a longer transfer.

```
udp-file-transfer-integrated-feature-metrics-llm-integration -- java -Dnettr...
echo "$RUNROOT" > /tmp/person4-runroot.txt
echo "NEW RUN: $RUNROOT"
NEW RUN: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-
integration/target/evaluation/person4-large-loss-2-20260927-032437
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-
integration % cd "$RUNROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=2 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=loss-2 \
-Dnettransfer.evaluation.record=true \
-cp "$JAR" nettransfer.Cli.TransferCliMain \
"$RUNROOT" \
['sample=storage/outgoing/sample.bin'
Simulator loss-2: receiver DATA loss=2.0%, DATA/ACK delivery delay=0 ms per direction,
seed=42; START/FINISH unaffected.
Evaluation records: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metri
cs-llm-integration/target/evaluation/person4-large-loss-2-20260927-032437/target/evalu
ation/llm-2026-09-26T23-25-29.7246922-15943919812862812801
Recorded model responses are untrusted evidence and may include answers rejected by Ja
va.
UDP transfer console. Type help for commands; natural-language requests use the config
ured interpreter.
transfer> start transfer {"file_id":"sample","receiver_id":"receiver-a","window_bytes"
:8192,"timeout_ms":500}
Start accepted [REAL]
transfer_id=dd070446-5a09-4e2f-909a-103113e4de39; run_id=dd070446-5a09-4e2f-909a-10311
3e4de39; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 m
s; retry limit: 5 consecutive rounds without progress
transfer> status last
```

Figure 14: Large-File Loss Transfer

```
person4-large-loss-2-20260927-032437 --zsh -- 82x27
NEW RUN: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-loss-2-20260927-032437
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration % cd "$RUNROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=2 \
-Dnettransfer.impairment.delayMs=0 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=loss-2 \
-Dnettransfer.logsRoot="$RUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] loss-2: receiver DATA loss=2.0%, DATA/ACK delivery delay=0 ms per direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-loss-2-20260927-032437/logs/standalone/run-64e6bb63-c80f-4377-b9b7-1c859b601c9f/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-loss-2-20260927-032437/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-large-loss-2-20260927-032437 %

person4-small-loss-2-20260927-032128 --java -Dnettransfer.impairment.enabled=true
3e4de39; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
Clarification: there is no completed or failed transfer in this session yet.
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:25:58.801111Z
transfer_id=dd070446-5a09-4e2f-909a-103113e4de39; run_id=dd070446-5a09-4e2f-909a-103113e4de39; protocol_transfer_id=bc697a29-9b70-477f-a19a-8edc321ab70b
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=2097275
sender_confirmed_acked_payload_bytes=2097275
sender_ack_progress_percent=100
live_elapsed_time_sec=18.1365975
sender_ack_based_rate_mbps=0.9251018555161739
packets_sent=2329
retransmissions=280
acks_received=2049
distinct_packets_acked=2049
packets_timed_out=35
rtt_sample_count=1780
rtt_mean_ms=1.5877648966292157
rtt_p95_ms=4.2355839999999999
endpoint_local_udp_payload_bytes_emitted=2447215
endpoint_emission_accounting_complete=true
transfer>
```

Figure 15: Large-File Loss Configuration

```
person4-large-loss-2-20260927-032437 --zsh -- 82x27
etrics-llm-integration/target/evaluation/person4-large-loss-2-20260927-032437/logs/standalone/reconciled/bc697a29-9b70-477f-a19a-8edc321ab70b/manifest.json
file_size_bytes: 2097275
payload_bytes_delivered: 2097275
transfer_time_sec: 18.1365975
throughput_mbps: 0.9251018555161739
transfer_success: True
integrity_verified: True
packets_sent: 2329
packets_received: 2287
acks_received: 2049
packets_timed_out: 35
packets_duplicated: 0
packets_dropped: 42
retransmissions: 280
retransmission_ratio: 0.12022327179046802
protocol_overhead_bytes: 405511
protocol_overhead_ratio: 0.1620238406319997
rtt_mean_ms: 1.5877648966292157
rtt_p95_ms: 4.2355839999999999
window_bytes_requested: 8192
window_packets: 8
timeout_ms: 500
packet_loss_rate: 2.0
delay_ms: 0.0
scenario: loss-2
(base) mozany@mozas-MacBook-Air person4-large-loss-2-20260927-032437 %

person4-small-loss-2-20260927-032128 --java -Dnettransfer.impairment.enabled=true
3e4de39; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
Clarification: there is no completed or failed transfer in this session yet.
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:25:58.801111Z
transfer_id=dd070446-5a09-4e2f-909a-103113e4de39; run_id=dd070446-5a09-4e2f-909a-103113e4de39; protocol_transfer_id=bc697a29-9b70-477f-a19a-8edc321ab70b
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=2097275
sender_confirmed_acked_payload_bytes=2097275
sender_ack_progress_percent=100
live_elapsed_time_sec=18.1365975
sender_ack_based_rate_mbps=0.9251018555161739
packets_sent=2329
retransmissions=280
acks_received=2049
distinct_packets_acked=2049
packets_timed_out=35
rtt_sample_count=1780
rtt_mean_ms=1.5877648966292157
rtt_p95_ms=4.2355839999999999
endpoint_local_udp_payload_bytes_emitted=2447215
endpoint_emission_accounting_complete=true
transfer>
```

Figure 16: Large-File Loss Results

The transfer still completed with verified integrity, but 42 packet drops and 280 retransmissions were recorded, while throughput decreased to 0.925 Mbps.

Delay Experiments

```

udp-file-transfer-integrated-feature-metrics-llm-integration -- java -Dnettr...
echo "$RUNROOT" > /tmp/person4-runroot.txt

echo "NEW RUN: $RUNROOT"
NEW RUN: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-
integration/target/evaluation/person4-small-delay-200-20260927-032812
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-i
ntegration % cd "$RUNROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=200 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=delay-200 \
-Dnettransfer.logsRoot="$RUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] delay-200: receiver DATA loss=0.0%, DATA/ACK delivery delay=200 ms pe
r direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature
-metrics-llm-integration/target/evaluation/person4-small-delay-200-20260927-032812
/logs/standalone/run-55de13cf-ff06-4260-b739-0428468c1913/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metr
ics-llm-integration/target/evaluation/person4-small-delay-200-20260927-032812/stor
age/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-small-delay-200-20260927-032812 %

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=200 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=delay-200 \
-Dnettransfer.logsRoot="$RUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] delay-200: receiver DATA loss=0.0%, DATA/ACK delivery delay=200 ms pe
r direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature
-metrics-llm-integration/target/evaluation/person4-small-delay-200-20260927-032812
/logs/standalone/run-55de13cf-ff06-4260-b739-0428468c1913/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metr
ics-llm-integration/target/evaluation/person4-small-delay-200-20260927-032812/stor
age/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-small-delay-200-20260927-032812 %

```

Figure 17: Small-File Delay Transfer

```

person4-small-delay-200-20260927-032812 -- -zsh -- 82x27
NEW RUN: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-
integration/target/evaluation/person4-small-delay-200-20260927-032812
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-i
ntegration % cd "$RUNROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=200 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=delay-200 \
-Dnettransfer.logsRoot="$RUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] delay-200: receiver DATA loss=0.0%, DATA/ACK delivery delay=200 ms pe
r direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature
-metrics-llm-integration/target/evaluation/person4-small-delay-200-20260927-032812
/logs/standalone/run-55de13cf-ff06-4260-b739-0428468c1913/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metr
ics-llm-integration/target/evaluation/person4-small-delay-200-20260927-032812/stor
age/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-small-delay-200-20260927-032812 %

500a274; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 m
s; retry limit: 5 consecutive rounds without progress
transfer> status last
Clarification: there is no completed or failed transfer in this session yet.
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:29:04.901469Z
transfer_id=83bf68d9-f736-4c70-b374-5bb2c500a274; run_id=83bf68d9-f736-4c70-b374-5bb2c
500a274; protocol_transfer_id=c9550002-c339-4b17-b375-6e3328b28159
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FI
NAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=13.512357
sender_ack_based_rate_mbps=0.1552753527752412
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=407.6396955291829
rtt_p95_ms=413.894333
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>

```

Figure 18: Small-File Delay Configuration

```

person4-small-delay-200-20260927-032812 -- -zsh -- 82x27
etrics-llm-integration/target/evaluation/person4-small-delay-200-20260927-032812/l
ogs/standalone/reconciled/c9550002-c339-4b17-b375-6e3328b28159/manifest.json
file_size_bytes: 262267
payload_bytes_delivered: 262267
transfer_time_sec: 13.512357
throughput_mbps: 0.1552753527752412
transfer_success: True
integrity_verified: True
packets_sent: 257
packets_received: 257
acks_received: 257
packets_timed_out: 0
packets_duplicated: 0
packets_dropped: 0
retransmissions: 0
retransmission_ratio: 0.0
protocol_overhead_bytes: 14462
protocol_overhead_ratio: 0.052260514799677665
rtt_mean_ms: 407.6396955291829
rtt_p95_ms: 413.894333
window_bytes_requested: 8192
window_packets: 8
timeout_ms: 500
packet_loss_rate: 0.0
delay_ms: 200.0
scenario: delay-200
(base) mozany@mozas-MacBook-Air person4-small-delay-200-20260927-032812 %

500a274; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 m
s; retry limit: 5 consecutive rounds without progress
transfer> status last
Clarification: there is no completed or failed transfer in this session yet.
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:29:04.901469Z
transfer_id=83bf68d9-f736-4c70-b374-5bb2c500a274; run_id=83bf68d9-f736-4c70-b374-5bb2c
500a274; protocol_transfer_id=c9550002-c339-4b17-b375-6e3328b28159
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FI
NAL; provenance=REAL
file_size_bytes=262267
sender_confirmed_acked_payload_bytes=262267
sender_ack_progress_percent=100
live_elapsed_time_sec=13.512357
sender_ack_based_rate_mbps=0.1552753527752412
packets_sent=257
retransmissions=0
acks_received=257
distinct_packets_acked=257
packets_timed_out=0
rtt_sample_count=257
rtt_mean_ms=407.6396955291829
rtt_p95_ms=413.894333
endpoint_local_udp_payload_bytes_emitted=269542
endpoint_emission_accounting_complete=true
transfer>

```

Figure 19: Small-File Delay Results

The next experiment introduced a 200 ms delivery delay with no intentional loss. For the small file, mean RTT increased to 407.64 ms; however, no timeout events or retransmissions occurred with the 500 ms timeout.

The delayed condition was then repeated with the large file using the same 8-packet window and 500 ms timeout.

```
udp-file-transfer-integrated-feature-metrics-llm-integration — java -Dnettr...
echo "$RUNROOT" > /tmp/person4-runroot.txt
echo "NEW RUN: $RUNROOT"
NEW RUN: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-20260927-033023
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration % cd "$RUNROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=200 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=delay-200 \
-Dnettransfer.logsRoot="$RUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] delay-200: receiver DATA loss=0.0%, DATA/ACK delivery delay=200 ms per direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-20260927-033023/logs/standalone/run-2dd9e2ee-5394-4cae-ac8c-92c9a8c791c8/events-receiver.jsonl

person4-small-delay-200-20260927-032812 — java -Dnettransfer.impairment...
java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=200 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=delay-200 \
-Dnettransfer.evaluation.record=true \
-cp "$JAR" nettransfer.cli.TransferCliMain \
"$RUNROOT" \
'sample-storage/outgoing/sample.bin'
Simulator delay-200: receiver DATA loss=0.0%, DATA/ACK delivery delay=200 ms per direction, seed=42; START/FINISH unaffected.
Evaluation records: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-20260927-033023/target/evaluation/llm-2026-09-26T23-30-55.247149Z-13452959727941564796
Recorded model responses are untrusted evidence and may include answers rejected by Java.
UDP transfer console. Type help for commands; natural-language requests use the configured interpreter.
transfer> start transfer {file_id:"sample",receiver_id:"receiver-a",window_bytes":8192,timeout_ms:500}
Start accepted [REAL]
transfer_id=71444d72-4ad0-4f76-9a46-0d0b6589a05c; run_id=71444d72-4ad0-4f76-9a46-0d0b6589a05c; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer>
```

Figure 20: Large-File Delay Transfer

```
person4-large-delay-200-20260927-033023 — zsh — 82x27
NEW RUN: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-20260927-033023
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration % cd "$RUNROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=200 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=delay-200 \
-Dnettransfer.logsRoot="$RUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] delay-200: receiver DATA loss=0.0%, DATA/ACK delivery delay=200 ms per direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-20260927-033023/logs/standalone/run-2dd9e2ee-5394-4cae-ac8c-92c9a8c791c8/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-20260927-033023/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-large-delay-200-20260927-033023 %

person4-small-delay-200-20260927-032812 — java -Dnettransfer.impairment...
Start accepted [REAL]
transfer_id=71444d72-4ad0-4f76-9a46-0d0b6589a05c; run_id=71444d72-4ad0-4f76-9a46-0d0b6589a05c; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:32:53.447503Z
transfer_id=71444d72-4ad0-4f76-9a46-0d0b6589a05c; run_id=71444d72-4ad0-4f76-9a46-0d0b6589a05c; protocol_transfer_id=dd68ac52-7b5a-4254-9945-ada41b0256d2
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=2097275
sender_confirmed_acked_payload_bytes=2097275
sender_ack_progress_percent=100
live_elapsed_time_sec=104.518224417
sender_ack_based_rate_mbps=0.16052894213988397
packets_sent=2049
retransmissions=0
acks_received=2049
distinct_packets_acked=2049
packets_timed_out=0
rtt_sample_count=2049
rtt_mean_ms=406.05438641922905
rtt_p95_ms=408.919
endpoint_local_udp_payload_bytes_emitted=2152935
endpoint_emission_accounting_complete=true
transfer>
```

Figure 21: Large-File Delay Configuration

```
person4-large-delay-200-20260927-033023 — zsh — 82x27
etrics-llm-integration/target/evaluation/person4-large-delay-200-20260927-033023/logs/standalone/reconciled/dd68ac52-7b5a-4254-9945-ada41b0256d2/manifest.json
file_size_bytes: 2097275
payload_bytes_delivered: 2097275
transfer_time_sec: 104.518224417
throughput_mbps: 0.16052894213988397
transfer_success: True
integrity_verified: True
packets_sent: 2049
packets_received: 2049
acks_received: 2049
packets_timed_out: 0
packets_duplicated: 0
packets_dropped: 0
retransmissions: 0
retransmission_ratio: 0.0
protocol_overhead_bytes: 111231
protocol_overhead_ratio: 0.0508364816758478356
rtt_mean_ms: 406.05438641922905
rtt_p95_ms: 408.919
window_bytes_requested: 8192
window_packets: 8
timeout_ms: 500
packet_loss_rate: 0.0
delay_ms: 200.0
scenario: delay-200
(base) mozany@mozas-MacBook-Air person4-large-delay-200-20260927-033023 %

person4-small-delay-200-20260927-032812 — java -Dnettransfer.impairment...
Start accepted [REAL]
transfer_id=71444d72-4ad0-4f76-9a46-0d0b6589a05c; run_id=71444d72-4ad0-4f76-9a46-0d0b6589a05c; protocol_transfer_id=unavailable
Requested window: 8192 bytes; effective window: 8 packets (8192 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:32:53.447503Z
transfer_id=71444d72-4ad0-4f76-9a46-0d0b6589a05c; run_id=71444d72-4ad0-4f76-9a46-0d0b6589a05c; protocol_transfer_id=dd68ac52-7b5a-4254-9945-ada41b0256d2
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=2097275
sender_confirmed_acked_payload_bytes=2097275
sender_ack_progress_percent=100
live_elapsed_time_sec=104.518224417
sender_ack_based_rate_mbps=0.16052894213988397
packets_sent=2049
retransmissions=0
acks_received=2049
distinct_packets_acked=2049
packets_timed_out=0
rtt_sample_count=2049
rtt_mean_ms=406.05438641922905
rtt_p95_ms=408.919
endpoint_local_udp_payload_bytes_emitted=2152935
endpoint_emission_accounting_complete=true
transfer>
```

Figure 22: Large-File Delay Results

The large-file delay experiment showed the same RTT behavior, with a mean RTT of 406.05 ms. The transfer required 104.52 seconds and achieved 0.161 Mbps while still completing without retransmissions or timeouts.

Window-Size Comparison

To examine the effect of window size under high latency, the large-file delay experiment was repeated with the window increased from 8 to 16 packets while all other parameters remained unchanged.

```
udp-file-transfer-integrated-feature-metrics-llm-integration -- java -Dnettr...
echo "$SRUNROOT" > /tmp/person4-runroot.txt
echo "FINAL RUN: $SRUNROOT"
FINAL RUN: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-window16-20260927-033446
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration % cd "$SRUNROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=200 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=delay-200-window16 \
-Dnettransfer.logsRoot="$SRUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] delay-200-window16: receiver DATA loss=0.0%, DATA/ACK delivery delay=200 ms per direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-window16-20260927-033446/logs/standalone/run-9b5c93acd-7661-4f81-9c25-4c2f5560744b/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-window16-20260927-033446/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-large-delay-200-window16-20260927-033446 %

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=200 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=delay-200-window16 \
-Dnettransfer.logsRoot="$SRUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] delay-200-window16: receiver DATA loss=0.0%, DATA/ACK delivery delay=200 ms per direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-window16-20260927-033446/logs/standalone/run-9b5c93acd-7661-4f81-9c25-4c2f5560744b/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-window16-20260927-033446/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-large-delay-200-window16-20260927-033446 %
```

Figure 23: 16-Packet Window Transfer

```
person4-large-delay-200-window16-20260927-033446 -- -zsh -- 82x27
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration % cd "$SRUNROOT"

java \
-Dnettransfer.impairment.enabled=true \
-Dnettransfer.impairment.lossPercent=0 \
-Dnettransfer.impairment.delayMs=200 \
-Dnettransfer.impairment.seed=42 \
-Dnettransfer.impairment.scenario=delay-200-window16 \
-Dnettransfer.logsRoot="$SRUNROOT/logs" \
-Dnettransfer.receiverInitialTimeoutMs=240000 \
-Dnettransfer.receiverInactivityTimeoutMs=15000 \
-jar "$JAR" receiver 9000 received.bin
[impairment] delay-200-window16: receiver DATA loss=0.0%, DATA/ACK delivery delay=200 ms per direction, seed=42; START/FINISH unaffected
[receiver] Binding UDP socket on port 9000 ...
[receiver] Waiting for a transfer...
[receiver] Event log: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-window16-20260927-033446/logs/standalone/run-9b5c93acd-7661-4f81-9c25-4c2f5560744b/events-receiver.jsonl
[receiver] Saved /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/target/evaluation/person4-large-delay-200-window16-20260927-033446/storage/incoming/received.bin
[receiver] SUCCESS: Transfer completed and verified
(base) mozany@mozas-MacBook-Air person4-large-delay-200-window16-20260927-033446 %

7ae056e; protocol_transfer_id=unavailable
Requested window: 16384 bytes; effective window: 16 packets (16384 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
Clarification: there is no completed or failed transfer in this session yet.
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:36:13.178786Z
transfer_id=d6188112-0e28-41ec-9e6c-b7fab7ee056e; run_id=d6188112-0e28-41ec-9e6c-b7fab7ee056e; protocol_transfer_id=8e05d63a-b4dc-4459-a2f9-647e6de9a4c1
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=2097275
sender_confirmed_acked_payload_bytes=2097275
sender_ack_progress_percent=100
live_elapsed_time_sec=52.553263125
sender_ack_based_rate_mbps=0.3192608603597534
packets_sent=2049
retransmissions=0
acks_received=2049
distinct_packets_acked=2049
packets_timed_out=0
rtt_sample_count=2049
rtt_mean_ms=406.3400082977064
rtt_p95_ms=409.999375
endpoint_local_udp_payload_bytes_emitted=2152935
endpoint_emission_accounting_complete=true
transfer>
```

Figure 24: 16-Packet Window Configuration

```
person4-large-delay-200-window16-20260927-033446 -- -zsh -- 82x27
-033446/logs/standalone/reconciled/8e05d63a-b4dc-4459-a2f9-647e6de9a4c1/manifest.json
file_size_bytes: 2097275
payload_bytes_delivered: 2097275
transfer_time_sec: 52.553263125
throughput_mbps: 0.3192608603597534
transfer_success: True
integrity_verified: True
packets_sent: 2049
packets_received: 2049
acks_received: 2049
packets_timed_out: 0
packets_duplicated: 0
packets_dropped: 0
retransmissions: 0
retransmission_ratio: 0.0
protocol_overhead_bytes: 111231
protocol_overhead_ratio: 0.050364816758478356
rtt_mean_ms: 406.3400082977064
rtt_p95_ms: 409.999375
window_bytes_requested: 16384
window_packets: 16
timeout_ms: 500
packet_loss_rate: 0.0
delay_ms: 200.0
scenario: delay-200-window16
(base) mozany@mozas-MacBook-Air person4-large-delay-200-window16-20260927-033446 %

7ae056e; protocol_transfer_id=unavailable
Requested window: 16384 bytes; effective window: 16 packets (16384 bytes); timeout: 500 ms; retry limit: 5 consecutive rounds without progress
transfer> status last
Clarification: there is no completed or failed transfer in this session yet.
transfer> status last
[REAL] COMPLETED; snapshot_at=2026-09-26T23:36:13.178786Z
transfer_id=d6188112-0e28-41ec-9e6c-b7fab7ee056e; run_id=d6188112-0e28-41ec-9e6c-b7fab7ee056e; protocol_transfer_id=8e05d63a-b4dc-4459-a2f9-647e6de9a4c1
engine_lifecycle=SUCCEEDED; evidence_scope=SENDER_ENDPOINT_LOCAL; evidence_finality=FINAL; provenance=REAL
file_size_bytes=2097275
sender_confirmed_acked_payload_bytes=2097275
sender_ack_progress_percent=100
live_elapsed_time_sec=52.553263125
sender_ack_based_rate_mbps=0.3192608603597534
packets_sent=2049
retransmissions=0
acks_received=2049
distinct_packets_acked=2049
packets_timed_out=0
rtt_sample_count=2049
rtt_mean_ms=406.3400082977064
rtt_p95_ms=409.999375
endpoint_local_udp_payload_bytes_emitted=2152935
endpoint_emission_accounting_complete=true
transfer>
```

Figure 25: 16-Packet Window Results

A final controlled experiment repeated the large-file delayed scenario while increasing only the window from 8 to 16 packets. Transfer time decreased from 104.52 to 52.55 seconds, while throughput increased from 0.161 to 0.319 Mbps. RTT remained approximately 406 ms, and no retransmissions or timeouts occurred.